

ITPROFEL2: OPEN SOURCE PROGRAMMING

Learning Module: Midterm - Week 3

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I. Overview and Objectives

Welcome to Week 3. Following our discussion on the philosophy and licensing of open source, we will now dissect the software ecosystem itself. We will categorize different types of open-source software, directly contrast them with proprietary equivalents, and explore the programming languages that power the open-source movement.

Intended Learning Outcomes (ILOs):

- Differentiate between Open Source Software and Proprietary Software.
- Identify various categories and real-world examples of Open Source Software.
- Define what makes a programming language "open source" and list the top languages used today.
- Map popular proprietary enterprise tools to their open-source alternatives.

II. Open Source vs. Proprietary Software

Software is broadly categorized by how its source code is distributed and legally restricted. Understanding this distinction is crucial for modern software engineers when deciding on a tech stack for a new project.

Feature	Open Source Software (OSS)	Proprietary Software (Closed Source)
Source Code	Freely available to the public to view, modify, and distribute.	Hidden and legally restricted. Only the creators can modify it.
Cost	Generally free to download and use (though support/hosting may cost money).	Usually requires a paid license, subscription, or upfront purchase.
Support	Relies heavily on community forums, wikis, and decentralized contributors.	Provided through official customer service channels and dedicated help desks.
Examples	Linux, Mozilla Firefox, VLC Media Player, WordPress.	Microsoft Windows, Adobe Photoshop, macOS.

III. Types of Open Source Software

Open source is not limited to just developer tools; it spans the entire spectrum of computing. Some major categories include:

- **Operating Systems:** Software that manages computer hardware and software resources (e.g., *Ubuntu, Fedora, Debian*).
- **Web Servers:** Software that serves web pages over the internet (e.g., *Apache HTTP Server, NGINX*).
- **Database Management Systems (DBMS):** Systems designed to manage, store, and retrieve data efficiently (e.g., *MySQL, PostgreSQL, MongoDB*).
- **Content Management Systems (CMS):** Platforms used to build and manage websites without needing to code from scratch (e.g., *WordPress, Joomla, Drupal*).

IV. Open Source Programming Languages

An open-source programming language is one where the compiler or interpreter, and often the standard library, is developed in the open. The community can contribute to the language's evolution, propose new features, and fix bugs.

Top Open Source Languages

- **Python:** Managed by the Python Software Foundation. Known for readability and heavily used in AI, data science, and web development.
- **JavaScript:** The backbone of the web. While ECMAScript is the standard, engines like V8 (used in Chrome and Node.js) are open source.
- **PHP:** A server-side scripting language specifically suited for web development. Powers a massive percentage of the internet, largely due to WordPress.
- **Go (Golang):** Developed by Google and open-sourced. Known for excellent concurrency and used heavily in cloud infrastructure (like Docker and Kubernetes).
- **Rust:** Managed by the Rust Foundation. Beloved for memory safety without garbage collection.

Student Activity: The Open Source Alternative Matrix

Instructions for Students:

For many businesses, transitioning to open source saves thousands of dollars in licensing fees. Your task is to act as an IT consultant for a startup company. Create a matrix (table) that maps the following expensive Proprietary Software to a viable Open Source Alternative. Briefly explain why the open-source alternative is a good replacement.

1. Microsoft Office Suite (Word, Excel, PowerPoint)
2. Adobe Photoshop
3. Adobe Illustrator
4. Microsoft SQL Server
5. AutoCAD

Prepare to present your matrix in our next laboratory session. Be ready to defend the viability of your open-source suggestions.